Subject Code 22ES1EE101	R22

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

B.Tech. I Year I Semester Regular Examinations, January 2024

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

(Common to CSE, IT, CyS and DS)

Time: 3 hours **Max. Marks: 60**

Answer ALL questions in PART-A.
Answer any ONE question from each unit in PART-B.
All Questions Carry Equal Marks.

PART-A

5X2=10M

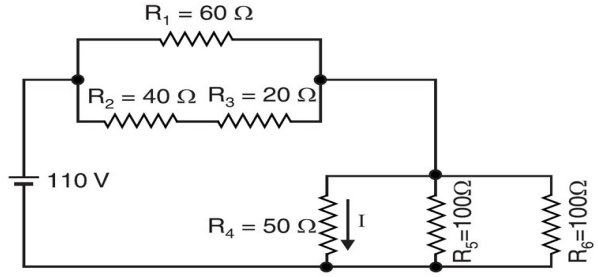
a)	Define KVL and KCL.	2M	CO-1	BL-1
b)	Draw the impedance triangle of series RLC Circuit.	2M	CO-2	BL-1
c)	List the speed control techniques of DC Motor.	2M	CO-3	BL-1
d)	Briefly explain the electrical characteristics of lead acid battery.	2M	CO-4	BL-1
e)	State PN junction diode and draw it's V-I Characteristics.	2M	CO-5	BL-1

PART- B

5X10=50M

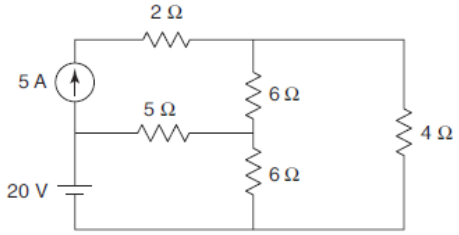
UNIT-1

1. Find the current I in the 50 Ω resistor in the circuit shown in figure. 10M CO-1 BL-3



OR

2. Find the current through the 4 Ω resistor by applying Superposition theorem. 10M CO-1 BL-3



UNIT-II

3. a) Derive the relation between Line Voltage and Phase Voltage in Star connected three phase system. 5M CO-2 BL-2
   b) Derive the relation between Line current and Phase current in Delta connected three phase system. 5M CO-2 BL-2

OR

4. A Voltage of 120 V at 50 Hz is applied to a resistance R in series with a capacitance C. The current drawn is 2 A and power loss in the resistance is 10M CO-2 BL-3

100 W. Calculate the (i) Resistance (ii) Capacitance and (iii) Power factor of the circuit.

UNIT-III

5.	Explain the Principle of Operation and Working of single -phase transformer with neat diagrams.	10M	CO3	BL-2
	OR			
6.	a) Derive the Expression for Generator EMF Equation.	5M	CO3	BL-2
	b) Explain the Working of DC Motor with neat diagrams	5M	CO3	BL-2

UNIT-IV

7.	a) Explain the principle of operation of Three phase Induction Motor.	5M	CO4	BL-2
	b) List the differences between Squirrel cage Induction Motor and slip ring induction motor.	5M	CO4	BL-2
	OR			
8.	a) With neat Diagrams explain the working of MCB.	5M	CO4	BL-2
	b) Define Earthing and Explain Its Types.	5M	CO4	BL-2

UNIT-V

9.	With neat Diagrams and Waveforms Explain the Operation of Full wave Rectifier.	10M	CO5	BL-2
	OR			
10.	Describe the working of NPN and PNP Transistors.	10M	CO5	BL-2

***